STRAY DOGS OVERPOPULATION IN MUNICIPALITY OF DAUIS: CELLULAR PHONE TRACKING TECHNOLOGY

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STRAY DOGS OVERPOPULATION IN MUNICIPALITY OF DAUIS: CELLULAR PHONE TRACKING TECHNOLOGY

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IMAGINATIVE ABSTRACT

Stray dogs have many negative impacts on the city environment, and human health, particularly in Dauis Bohol. Stray dogs can cause a collision when they run across the road, injuring other people and themselves. The purpose of this study is to take advantage of the advanced technology in implementing this scheme by employing a prototype collar with a GPS (Geographic-Positioning System) mobile enhancer for a tracking device. To develop cellular phone tracking, providing the three (3) sections: the location service section for accessing information, GPS (Geographic-Positioning System) for recording the exact dog locations, and a prototype collar device attached to stray dogs. In this project, we are using ARDUINO UNO, GPS BOARD, ACCELEROMETER, BATTERY, SD CARD, and a COLLAR for identification tags with the prototype is attached. The GPS Tracking Device will be placed in a 3D printed plastic housing installed in the standard collar worn on stray dogs' neck. The stray dogs captured in every barangay will be examined and will undergo a veterinary checkup. Hence, they will be identified whether they need a collar or not. The device is programmed using Arduino's Repositories based on the C++ language, and we will be using a website called "GPS VISUALIZER" to convert the file. Geographic-Positioning Systems (GPS) is more important to make it easier to track pets, specifically for stray dogs.

CCS Concepts

• Computer System Organization → Real-time System

• Hardware → Communication hardware, Interfaces and storage → Wireless Device

• Networks → Network types → Wireless Access Networks

Keywords

Stray Dogs; Tracking Technology; GPS; Prototype Collar Device; Cellular Phone

1. INTRODUCTION

According to Beck 2013, humans' close association with animals such as dogs and cats has substantial positive benefits. These positive benefits could be psychological and physiological [10]. The bond between humans and animals is becoming impenetrable since both can correlate in many ways [2]. The presence of companion animals plays various roles in our everyday life. Humans, therefore, must protect animals since man cannot do without them. Nowadays, billions of dogs of different breeds share our space as close companions or as stray or feral ones. With the worsening situation, breeding among strayed animals can't be controlled. As a result, the population of stray animals continues growing in the number. With advanced technology, solutions to end the overpopulation of stray pets have been urged using a tracking system.

In the 1950s and 1960s, Radio Telemetry revolutionized the study of animal, in enabling routine, systematic measurements of animal's location that emits very high frequency (VHF) radio signal pulses to their tags or collars and has been successfully used since the beginning (Cochran and Lord 1963, Le Munyan et al., 1959) [4][5]. Nevertheless, VHF system usage was expensive, long-winded, and often risky to personnel safety [6]. Over time, Nimbus 3 satellite launched (Kenward's 1987) [8] and, later, Argos System (Fancy et al. 1988) [7], collecting and transmitting location data from a wide range of animals become possible with the usage of satellite communication technology [2]. Besides, the location's position accuracy quotes coarse (Britten et al., 1999) [9]. The first advent of the Geographic- Positioning System (GPS), having a remarkable position accuracy announced as fully operational in 1955 (Rodger et al. 1996) [3].

With the fast-growing development in the Municipality of Dauis, stray dogs have become one of the significant problems of public management in the lives of Dauisanon and a widespread concern by the public. Statistics resulting from the Provincial Vet Office in Tagbilaran City showed that out of the 47 towns and one city in Bohol, Dauis are among others that don't have its Dog Shelters. The research objective is to implement the definite fields of the above study development of a Cellular phone Tracking Device. The Location Service Section provides three sections for accessing information, GPS Satellite for positioning, and a Prototype Collar Device. The location data fixed by GPS positioning are upload to the location information server. Operators need to access the location server to acquire the stored location date or a current location. The collars will be transmitting a corrected GPS location within the range. Collar location data received will be stored and transferred to the location information server.

Possess the same target with the BARK, INC. (Bohol Animal Rescue and Kindness), an animal welfare organization dedicated to helping animals in need, solving stray population through their CNVRM (Catch, Neuter, Vaccine, Release, Manage) project, education, and adoption. We conduct this study working with the BARK, INC. to bring solutions to loads of negative impacts in the environment and human health caused by stray dogs and considers the responsibility to help and value their rights to live. Collecting Data on the numbers of stray dogs will be undertaken by their expertise and experience with our proposed model. These areas take up a study of great importance to contribute an improved understanding of our stray Dog's behaviour, eliminate the risk caused by our gets lost dogs, and reserve the dimensions underlying the dog-human relationship.

2. PROPOSED METHODOLOGY

2.1 Study Area

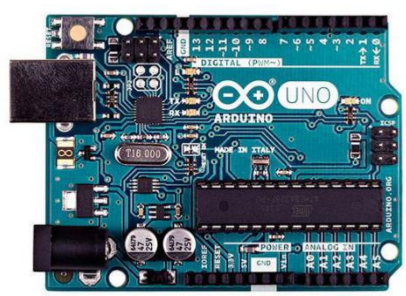
The research study holds in the municipality of Dauis in the coastal town located in northern parts of Panglao Island to the southwestern part of Bohol, 12km away from Tagbilaran City (9.6074° N, 123.8234° E). Dauis is specified as the main area of our study due to an ever-worsening stray dogs' problem and to make an alternative to avoid shoot-to-kill order. The municipality has a homeland area of 43.33 square kilometres or equalled to 16.73 square miles to constitute 0.90% of Bohol's total area. Its population determined by the 2015 Census was 45,663 total in 12 barangays. The data represented 3.48% of the total population of Bohol province.

2.2 Capturing, Instrumenting, and Releasing Stray Dogs

Cooperating with the BARK, INC., the planed, and techniques from capturing to releasing will be guided by their knowledge and skills. Trapping of at least one stray dogs per barangay area, a total of 12 stray dogs, will be administered to be studied first. The stray dogs captured in every barangay will be examined and undergo veterinary supervision before handling the captive. The stray dogs that have risky diseases will not be qualified for the selection of the study. Keeping a record of Biometric Parameters for each stray Dog is beneficial for tracking identity and the collars' fitting. Because this research is collaborating into a CVNRM Program, Capture, Neuter, and Release will be the basis of holding the study. Stray male dogs will be castrated, and stray female dogs will undergo spaying regarding the program's implementation. This method will be done before fitting and releasing to control stray dogs' breeding.

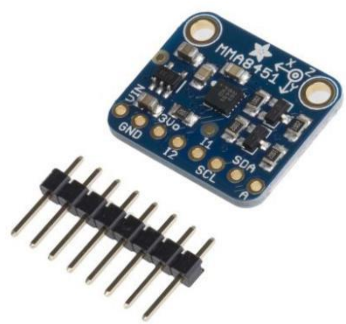
2.3 HARDWARE DESCRIPTION

Arduino UNO is an open-source electronics and programming platform that can read inputs and turn them into an output.



GPS

board (ADAFRUIT Ultimate GPS + Logging Shield) - A GPS board designed to log data to an SD card.

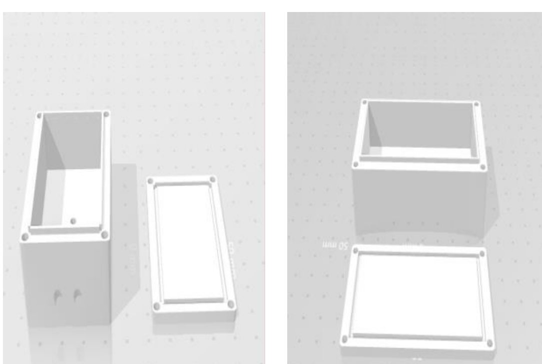


Accelerometer - A motion sensor device.

Battery (Lithium-Ion 4400 MAH Rechargeable) - is a device that supplies electric power.

Micro SD Card - removable flash memory card used to store files.

Collar (standard dog collar w/ metal buckle) is a detachable piece used for Identification tags that the prototype is attached.



3D printed plastic Housing (clear polyester resin) - used for the casing.

Figure 1. Block Diagram of GPS Tracker:

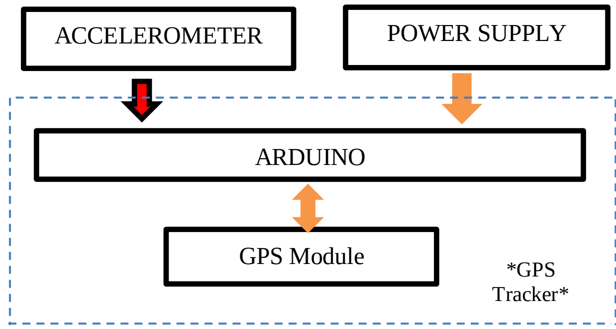
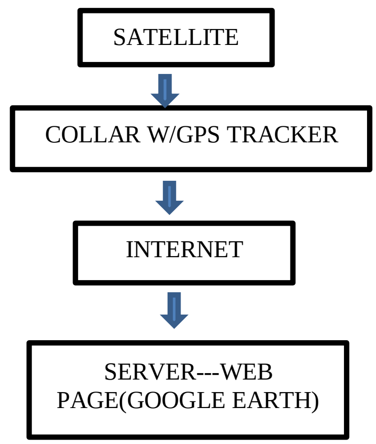


Figure 2. Representation of GPS Tracking Operation:



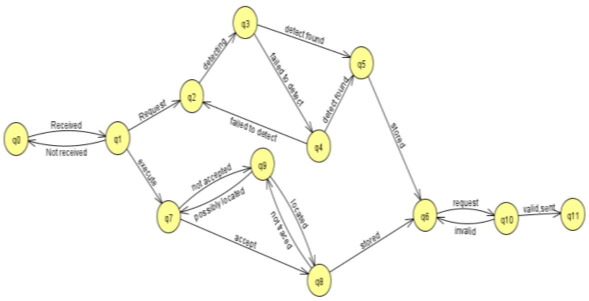
2.4 Hardware Implementation

The GPS Tracker Device will be put in a 3D printed plastic housing and installed on a standard collar. It's paste at the neck part of the stray dogs.

2.5 Software Implementation

The machine is programmed based on the C++ language using the Repositories of Arduino. The programming lets the frequency of data collection be replaced by altering parameters in the setup code. Whenever the Accelerometer does not detect movement, the program executes to sleep for a lifesaver battery. We will use an existing application to map the data, which we choose Google Earth for our web page. Fortunately, the TXT extension format file created by the GPX needs conversion from its original format accepted by Google earth with the GPX extension in using a website called "GPS visualizer" to convert the file to carry out.

Figure 3. Automata of GPS tracker



State:

q0 Aerial

q1 GPS Validation State

q2 Accelerometer State

q3 Movement Detection

q4 Non-Movement Detection

q7 Body Identification State

q8 Tracker State

q9 Body Detector State

q10 Transmission State

q11 Stored on System Network

Output State:

q5 Motion Filter (Accelerometer)

q6 Memory storage (data logger)

Figure 4: The Arduino Repositories Code

```
//Code for arduino GPS collars using Adafruit GPS
and accelerometer
#include <SPI.h>
#include <Adafruit_GPS.h>
#include <TinyGPS.h>
#include <SD.h>
//#include <SoftwareSerial.h>

//package for making the microcontroller sleep,
 might be useful if accelerometer data is not
 required just used for smart collars
//#include <avr/sleep.h>

#include <Wire.h>
#include <Adafruit_LIS3DH.h>
#include <Adafruit_Sensor.h>

// define the pin that will turn on and off the GPS
to save power High is off Low is on
// #define Enable 6

//not sure these are strictly necessary as with this
wiring the accelerometer is running I2C but if wired
different will need one of these
// Used for software SPI
//#define LIS3DH_CLK 13
//#define LIS3DH_MISO 12
//#define LIS3DH_MOSI 11
// Used for hardware & software SPI
//#define LIS3DH_CS 10

//assign the accelerometer
Adafruit_LIS3DH lis = Adafruit_LIS3DH();

// Set GPSECHO to 'false' to turn off echoing the
 GPS data to the Serial console
// Set to 'true' if you want to debug and listen
to the raw GPS sentences
#define GPSECHO  true
/* set to true to only log to SD when GPS has a fix,
 for debugging, keep it false */
#define LOG_FIXONLY false

// define the GPS (with the serial)
Adafruit_GPS GPS(&Serial1);

// set up the files that will be used, one to save
 accelerometer data one for gps data
File accellog;
File GPSlog;

//Assign time variable currently just tracking
miliseconds since it turned on
unsigned long time;

//Assign File Name Variables
char filename1[15];
char filename2[15];

//create a class for Accelerometer readings so multiple
 can be stored across different times and then logged
 as a batch to save battery
class Accelerometer
  {
    //class member variables

    // read is an array which means it has a length
    of this much, so you will store it into the array,
    once the array is full log it,
    // curre ntly set to 100 which is 10 seconds
    public:
    float xRead[100];
    float yRead[100];
    float zRead[100];

     long interval;
     unsigned long previoustime;
     unsigned long Time[100];
     float x0;
     float x100;
   //the public values of this

   Accelerometer(long Interval)
   {
     Interval = interval;

     memset(xRead,0,sizeof(xRead));
     memset(yRead,0,sizeof(yRead));
     memset(zRead,0,sizeof(zRead));
     memset(Time,0,sizeof(Time));

     previoustime = 0;

     x0 = xRead[0];
     x100 = xRead[100];


   }

//create a function to read the accelerometer
void readaccelerometer()
{
   unsigned long currentMillis = millis();
   sensors_event_t event;
   lis.getEvent(&event);
    int i = 0;
   if(currentMillis - previoustime >= interval)
   {
     xRead[i] = event.acceleration.x;
     previoustime = currentMillis;

     yRead[i] = event.acceleration.y;
     previoustime = currentMillis;


   zRead[i] = event.acceleration.z;
   previoustime = currentMillis;

   Time[i] = millis();

   i = i+1;
  }

  if(i = 100)
  {
       accellog = SD.open(filename1, FILE_WRITE);
   for (int j = 0; j<100; j++)
   // writing the accelerometer data to the SD card
   {
     if (accellog) {
   accellog.print("Time: "); accellog.print(Time[j]);accellog.print("  ");
   accellog.print("X: "); accellog.print(xRead[j]); accellog.print("  ");
   accellog.print("Y: "); accellog.print(yRead[j]); accellog.print("  ");
   accellog.print("Z: " ); accellog.print(zRead[j]); accellog.print("  ");
   }
   }
   accellog.close();
   i=0;
  }

 }

};

//A class for the GPS
class GPSchip
{
  //class member variables
  int Offpin;
  long interval;
  int PinState;
```

```
  int attempts;
  long fixinterval;
  unsigned long previoustime;
  boolean On;

  public:
  // the variables entered into the GPS when constructed,
  interval is time between fixes in minutes (alter equation if you want it in seconds),
  // Enable is the pin number that the enable pin is soldered to ,
  // Checkfix is how many minutes you want it to search for
  a fix before it turns off
  GPSchip(int Interval, int Enable, int Checkfix)
  {
    interval = ((Interval * 60) / 100);
    Enable = Offpin;
    PinState = HIGH;
    attempts = 0;
    previoustime = 0;
    fixinterval = ((Checkfix * 60) / 1000);
    On = false;
    pinMode(Enable, OUTPUT);

  }

void GPSread()
{

  unsigned long currentMillis = millis();
  unsigned long Starttime;
  if(currentMillis - previoustime >= interval)
  {
    digitalWrite(Offpin,LOW);

    GPS.sendCommand(PMTK_SET_NMEA_OUTPUT_RMCGGA);
    GPS.sendCommand(PMTK_SET_NMEA_UPDATE_1HZ);
    GPS.sendCommand(PGCMD_NOANTENNA);

    Starttime = currentMillis;

    On = true;
    previoustime = currentMillis;
  }

  if (On && LOG_FIXONLY && GPS.fix)
    {
     if (GPSlog) {
    GPSlog.print("Time: "); GPSlog.print(time); GPSlog.print("  ");
    GPSlog.print("GPS: "); GPSlog.println(GPS.read());
    digitalWrite(Offpin, HIGH);
    On = false;
    }
    }
   if (currentMillis - Starttime >= fixinterval && !GPS.fix)
    {
    digitalWrite(Offpin, HIGH);
    On = false;
    }
  }

};

void setup() {
// put your setup code here, to run once:

// this sets the filename and checks if it already exists on
the card, if it does it changes the name  to something else maybe?
strcpy(filename1, "ACCLOG00.TXT");
  for (uint8_t i = 0; i < 100; i++) {
    filename1[6] = '0' + i/10;
    filename1[7] = '0' + i%10;
    // create if does not exist, do not open existing, write,
    sync after write
    if (! SD.exists(filename1)) {
      break;}}

strcpy(filename2, "GPSLOG00.TXT");
  for (uint8_t i = 0; i < 100; i++) {
    filename2[6] = '0' + i/10;
    filename2[7] = '0' + i%10;
    // create if does not exist, do not open existing, write,
    sync after write
    if (! SD.exists(filename2)) {
      break;}}


  }
void loop() {
  // put your main code here, to run repeatedly:

Accelerometer Accel(100);
GPSchip GPS1(5,6,2);

Accel.readaccelerometer();

if ((Accel.x0 - Accel.x100) != 0)
  {
    GPS1.GPSread();
  }


}
// end code
```

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